MIL-T-462E

16 September 1985

SUPERSEDING
MIL-T-462D
23 March 1978

MILITARY SPECIFICATION

TANKS, LIQUID STORAGE, METAL:

FUEL-OIL, HORIZONTAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers horizontal, nonpressure-type, fuel-oil, storage tanks.

1.2 Classification. The storage tanks shall be of the following types and sizes, as specified (see 6.2):

Type II - Underground storage tanks.

Sizes: 275 gallon nominal capacity (1041 liters).
550 gallon nominal capacity (2082 liters).
1,000 gallon nominal capacity (3785 liters).
5,000 gallon nominal capacity (18925 liters).

Type III - Above ground storage tank.

Size: 275 gallon nominal capacity (1041 liters).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5430

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-C-494	- Coating Compound, Bituminous, Solvent Type, Acid Resistant.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-T-704	- Treatment and Painting of Materiel.
DOD-P-15328	- Primer (Wash), Pretreatment (Formula No. 117 for Metals) (Metric).
MIL-C-46168	- Coating, Aliphatic Poluyrethane, Chemical Agent Resistant.
MIL-P-52192	- Primer Coating, Epoxy.
MIL-C-52950	- Crates, Wood, Open and Covered.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practices for Commercial Packaging.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

ANSI/UL 58 - Standard for Steel Underground Tanks for Flammable and Combustible Liquids.
ANSI/UL 80 - Standard for Steel Inside Tanks for Oil-Burner Fuel.

(Application for copies should be addressed to American National Standards Institute, Inc., 1430 Broadway, New York, New York 10018.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA No. 30 - Flammable and Combustible Liquids Code.

(Application for copies should be addressed to the National Fire Protection Association, Battery March Park, Quincy, MA 02269.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The fuel-oil storage tanks hereinafter referred to as "tanks", shall be approved for the purpose intended by Underwriters Laboratories, Inc. (UL) or the contractor may submit independent proof, satisfactory to the contracting agency, that the tanks conform to the applicable requirements of the published standards including methods of test of UL.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

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3.3 Materials. Materials shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Materiel deterioration prevention and control. The tanks shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operation and storage environment to which the tanks may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and subcomponents, and shall make the information available, upon request, to the contracting officer or designated representative.

3.3.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the tanks may be newly fabricated from recovered materials to the maximum extent practicable, provided the tanks produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the tanks.

3.4 Construction.

3.4.1 Openings. Unless otherwise specified (see 6.2), all threaded openings shall be of the size and location as shown in Figure 1 and shall be in accordance with the applicable standard of the Underwriters Laboratories, Inc. as specified herein. When specified (see 6.2), the tanks shall be equipped with access manholes fitted with bolted removable covers for cleaning and inspection purposes.

3.4.2 Type II underground storage tanks. Type II tanks shall be fabricated in accordance with Underwriters Laboratories, Inc., ANSI/UL 58 or the applicable requirements of published standards.

3.4.3 Type III, 275-gallon-capacity, above ground storage tanks. Type III tanks shall be fabricated in accordance with Underwriters Laboratories, Inc., ANSI/UL 80 or the applicable requirements of published standards.

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3.4.3.1 Mounting brackets. Two 4 by 4 by 0.250 inch (102 by 102 by 6.4 millimeters (mm)) steel angle brackets 4 inches (102 mm) long shall be welded, with a continuous full fillet weld, on each side of the tank, symmetrically located approximately one-fourth the length of the tank from each end and 2 inches (51 mm) up from the bottom. Each bracket shall be drilled and tapped for a 1.500 inch (38 mm) pipe leg.

3.5 Human factor engineering. The horizontal fuel oil storage tanks shall conform to accepted human factor engineering criteria as described in MIL-STD-1472.

3.6 Safety. Type III tanks shall have an emergency relief venting capacity in accordance with NFPA 30 to relieve excessive internal pressure caused by exposure fires.

3.7 Treatment and painting.

3.7.1 Treatment and coating. Exterior surfaces of type II tanks shall be cleaned in accordance with MIL-T-704 and pretreated with wash primer conforming to DOD-P-15328. The tanks shall then be given a coat of epoxy primer conforming to MIL-P-52192 to a minimum 1 mil dry film thickness, and then coated with compound, bituminous solvent type, acid resistant, conforming to TT-C-494, for a total thickness of not less than 0.0625 inch (1.6 mm).

3.7.2 Treatment and painting. The portions of the type III tanks normally painted internally and externally shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G, as applicable. Unless otherwise specified (see 6.2) top coat color shall be camouflage green 383 conforming to MIL-C-46168.

3.8 Identification marking. The tanks shall be identified in accordance with MIL-STD-130.

3.8.1 Safety marking. The type III tanks shall be stenciled in 2 inch (50.8 mm) lusterles red letters on both sides of the tank "FLAMMABLE-KEEP FIRE AWAY".

3.9 Workmanship. All parts of the tanks shall be free of harmful extraneous material such as sand, dirt, pits, sprues, scale and flux. Metal used in fabrication shall be free from kinks and sharp bends. All bends shall be made with controlled means to insure uniformity of size and shape. The surface of parts to be welded shall be free from rust, scale, paint, grease, undercuts, and foreign matter. Boltholes shall be accurately formed and all burrs removed.

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4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 First article examination. The first article shall be examined as specified in 4.5.1, in the order shown in table I. The presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.5 Examination procedure.

4.5.1 Examination. The tanks shall be examined as specified herein for the following defects:

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TABLE I. Examination schedule.

Number	Defects	Requirement Paragraph
Major		
101.	Underwriters label indicating approval under ANSI/UL 58 or ANSI/UL 80, as applicable, not attached to the tank.	3.1
102.	Material not as specified.	3.3
103.	Materials not resistant to corrosion and deterioration or treated to be made resistant to corrosion and deterioration.	3.3.1
104.	Dissimilar metals not in accordance with MIL-STD-889.	3.3.2
105.	Material and material finish or treatment information not available.	3.3.3
106.	Rebuilt or remanufactured components are incorporated in the tank.	3.3.4
107.	Openings not as specified.	3.4.1
108.	Mounting brackets not as specified.	3.4.3.1
109.	Human factors not as specified.	3.5
110.	Type III tanks does not have an emergency relief venting capacity as specified.	3.6
111.	Treatment and coating not as specified for type II tank.	3.7.1
112.	Treatment and painting not as specified for type III tank.	3.7.2
113.	Paint color not as specified.	3.7.2
114.	Identification marking not as specified.	3.8
115.	Safety marking not as specified.	3.8.1
116.	Workmanship not as specified in ANSI/UL 58, ANSI/UL 80 or the requirements of published standards.	3.9

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a complete tank prepared for shipment shall be considered a unit of product.

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4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

No.	Defect	A	Comm.
117.	Preservative application and selection not as specified.	5.1.1.1	5.1.2
118.	Interior surfaces of tank not preserved as specified, when required.	5.1.1.2	
119.	When required, excess preservative not drained as specified.	5.1.1.2	
120.	Openings not sealed as specified.	5.1.1.2	
121.	Tanks not packed as specified.	5.2.1.1/5.2.1.2	5.2.2
122.	Marking not as specified.	5.3.1	5.3.2

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Preservative application and selection. The provisions of MIL-P-116 shall apply when selecting and determining the applicability of preservatives. Preservatives specified herein by "P" numbers shall conform to the applicable specifications listed in MIL-P-116 and shall be applied in accordance with MIL-P-116.

5.1.1.2 Interior surfaces. Interior surfaces of tanks requiring a contact preservative (see 5.1.1.1), shall be coated with type P10, type I, grade 30 preservative. Excess preservative shall be drained, and all openings shall be sealed with metal or plastic caps or plugs as applicable.

5.1.2 Commercial. The tanks shall be preserved in accordance with ASTM D 3951.

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5.2 Packing. Packing shall be level A or commercial as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 275- and 550-gallon tanks. Type II, sizes 275- and 550-gallon tanks, and type III, 275-gallon tanks, preserved as specified in 5.1, shall be packed in crates conforming to MIL-C-52950, type III, style B. Anchoring, cushioning, closure and strapping shall be as specified in the appendix thereto.

5.2.1.2 1000- and 5000-gallon tanks. Type II, sizes 1000- and 5000- gallon tanks, preserved as specified in 5.1, shall not be crated.

5.2.2 Commercial. Tanks of all types and sizes preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Marking.

5.3.1 Level A. Level A marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Commercial marking shall be in accordance with ASTM D 3951. Additionally, each tank or shipping container shall be marked with the gross weight and cube.

6. NOTES

6.1 Intended use. The tanks are intended for use in fuel-oil storage in connection with oil-burner installations. Type III tanks are intended for installation indoors, although outdoor installations are not precluded.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, and size of tank required (see 1.2).

c. When a first article is required for inspection and approval, and the number of units required (see 3.2).

d. Threaded openings required if other than those specified in 3.4.1.

e. When manhole is required (see 3.4.1).

f. Color required when other than as specified (see 3.7.2).

g. Degree of preservation and packing required (see 5.1 and 5.2).

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6.3 First article. When a first article inspection is required, the items should be a tank(s). The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests and approval of the first article test results and disposition of the documents first articles.

6.4 Information figure. Figure 1 shows the approximate overall dimensions of tanks which have been found acceptable; however, the figure is included for illustration only and is not intended to preclude the furnishing of other overall tank dimensions which conform to this specification.

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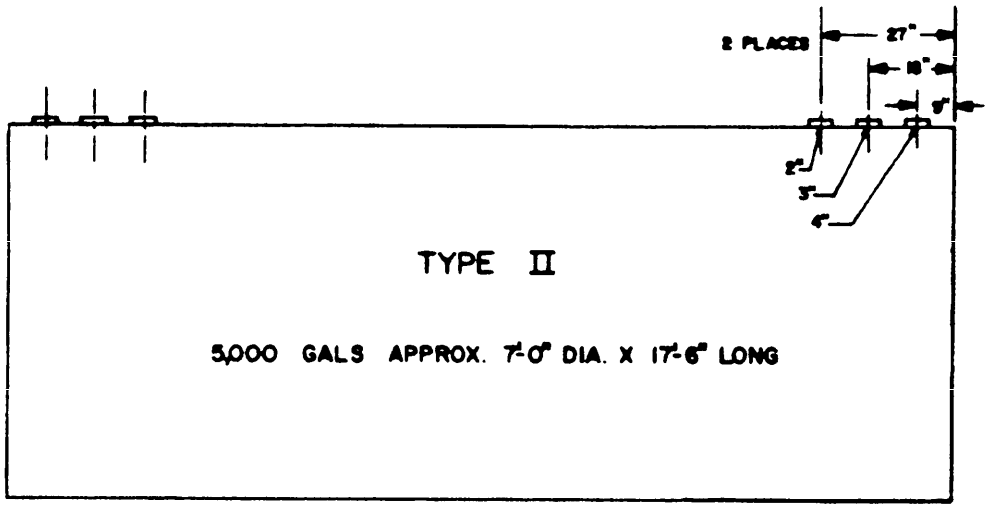
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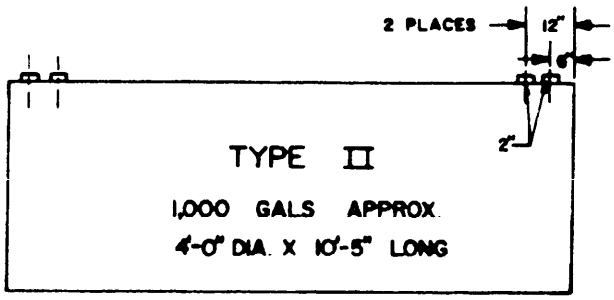
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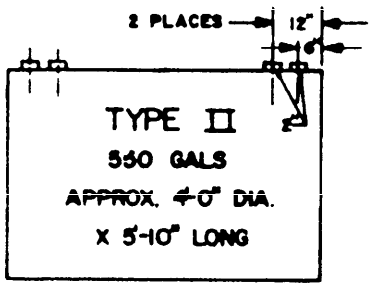
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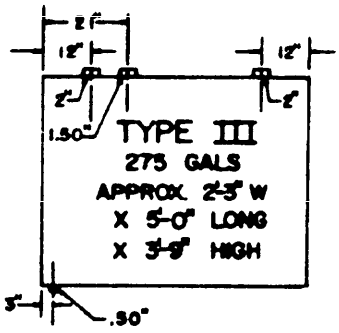


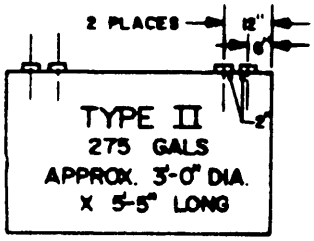


DIMENSIONS			
INCHES	MM	FEET AND INCHES	MM
.50	12.7	2' - 3"	685.8
1.50	38.1	3' - 0"	914.4
2	50.8	3' - 9"	1 143.0
3	76.2	4' - 0"	1 219.2
4	101.6	5' - 0"	1 524.0
6	152.4	5' - 5"	1 651.0
9	228.6	5' - 10"	1 778.0
12	304.8	7' - 0"	2 133.6
18	457.2	10' - 5"	3 175.0
21	533.4	17' - 6"	5 334.0
27	685.8		



CAPACITIES	
GALLONS	LITERS
275	1 041
550	2 082
1,000	3 785
5,000	18 925





NOTES:

1. ALL OPENINGS INTERNALLY THREADED NPT.
2. TOLERANCE ON DIMENSIONS LOCATING OPPENINGS IS±.50 INCH.

FIGURE 1. Location and size of tank openings.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-462E	Tanks, Liquid Storage, Metal: Fuel-Oil, Horizontal

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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